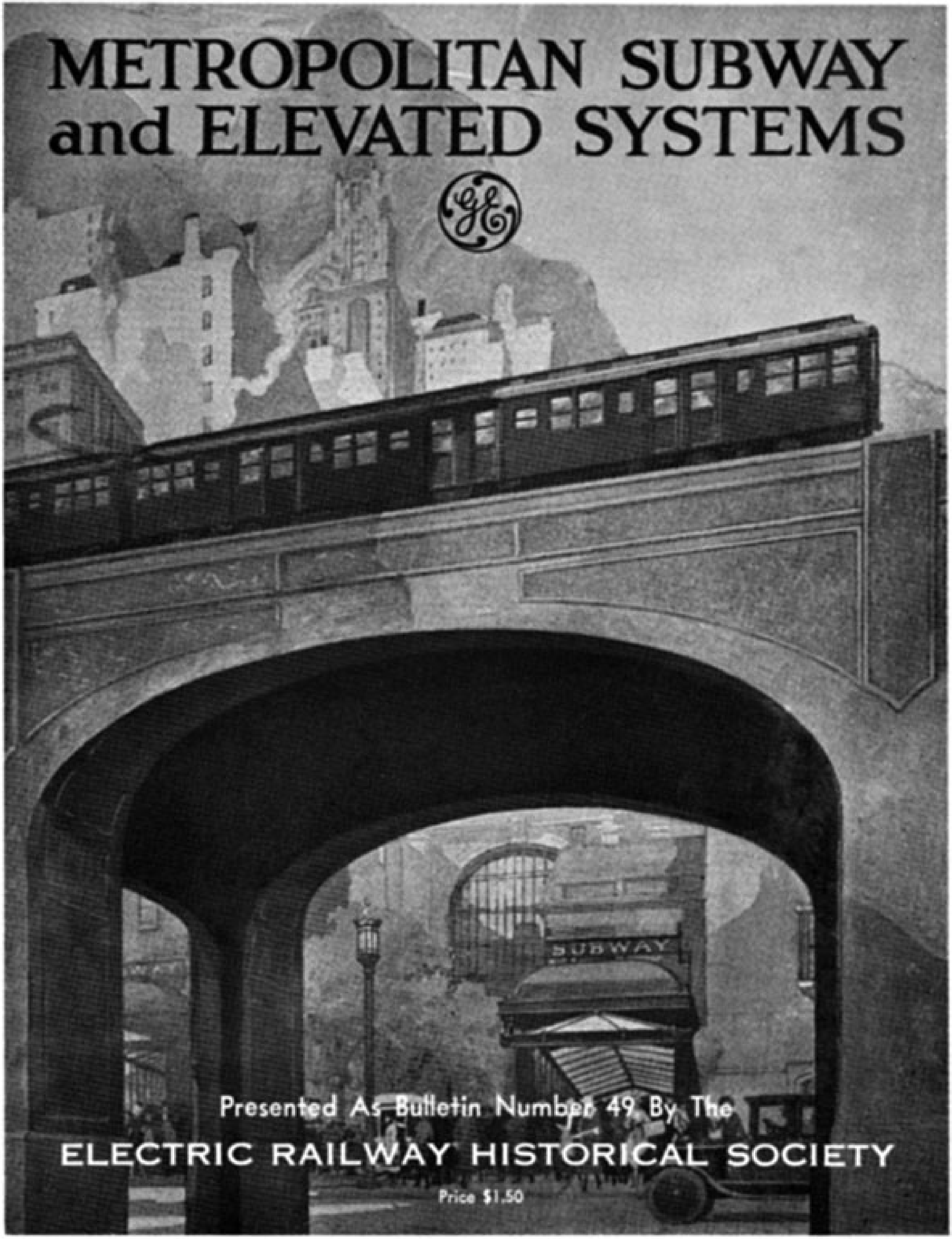
METROPOLITAN SUBWAY and ELEVATED SYSTEMS



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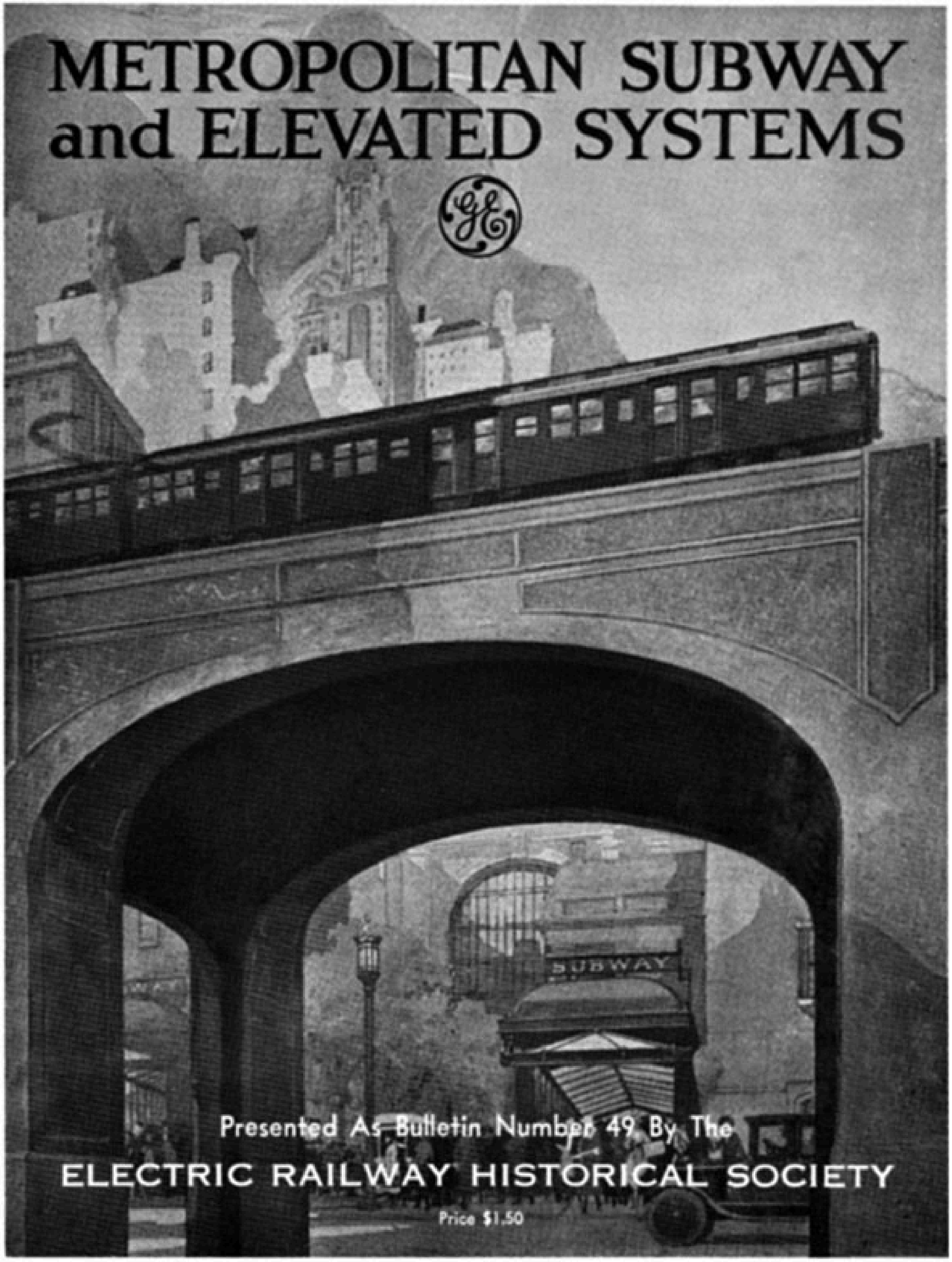

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Transcriber Notes

- Obvious typos and punctuation errors fixed.

- Inconsistencies in hyphenation kept as in the original.

METROPOLITAN SUBWAY
and ELEVATED SYSTEMS

GE

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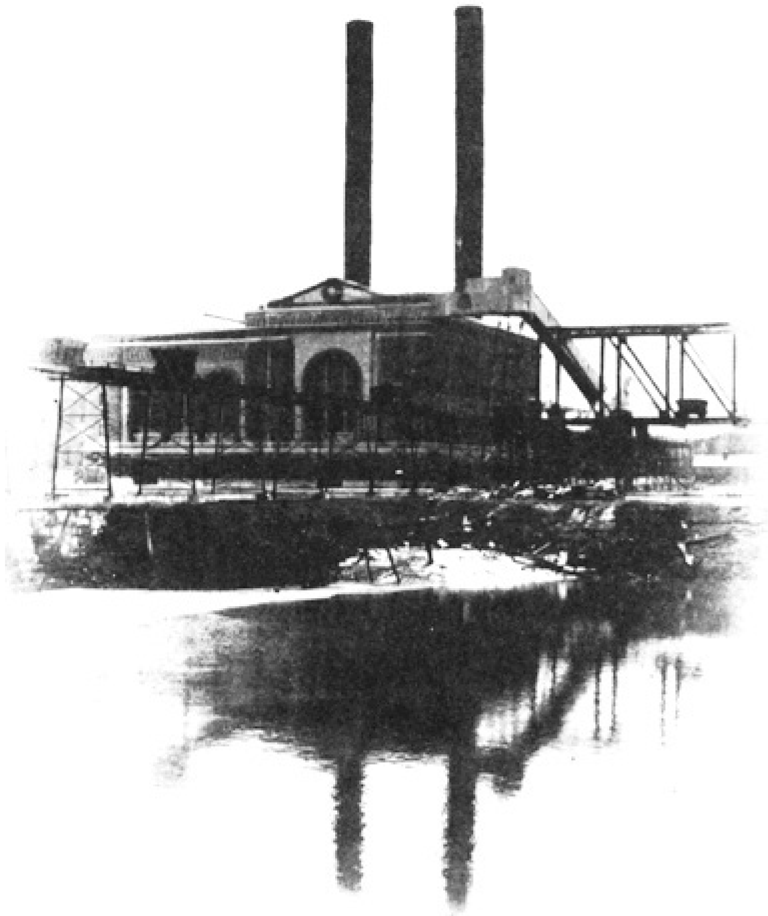
BOSTON ELEVATED RAILWAY RAPID TRANSIT SYSTEM

The transportation system of the city of Boston comprises a combination of both rapid transit and surface lines operated under a single fare arrangement with transfer privileges permitting a continuous ride in one general direction from one end to the other of the system. The elevated lines and the Tremont St. Subway were originally built by the railway company in 1901. Today the total transportation system includes more than 500 miles of line of which 37 miles are subway and elevated tracks. The population served in the district of more than 92 square miles is considerably over a million people and the number of revenue passengers carried, approximates 350,000,000 per year. Statistics are not available as to the passengers carried on the Rapid Transit lines.

The original elevated structure operated between Sullivan Sq., Charlestown, and Dudley Street, with two branches through the city, one by subway under Tremont St. and the other by the way of Atlantic Ave. and South Station. In 1908-9 the elevated structure was extended to the present terminal at Forest Hills and the Washington St. Subway was completed through the business part of the city. The Cambridge Subway was placed in operation in 1912. Recent extensions include an elevated line from Sullivan Square to Everett and reconstruction of the tunnel to East Boston.

Since July 1, 1919, the system has been operated by the Board of Trustees of the Commonwealth. Under the direction of this board are included not only the details of operation and management, but also the decisions as to fares to be charged independent of the State Department of Public Utilities.

Under the direction of the present management a continuous program of improvements has been inaugurated which has necessitated the re-routing of trains to handle the traffic to the best advantage.

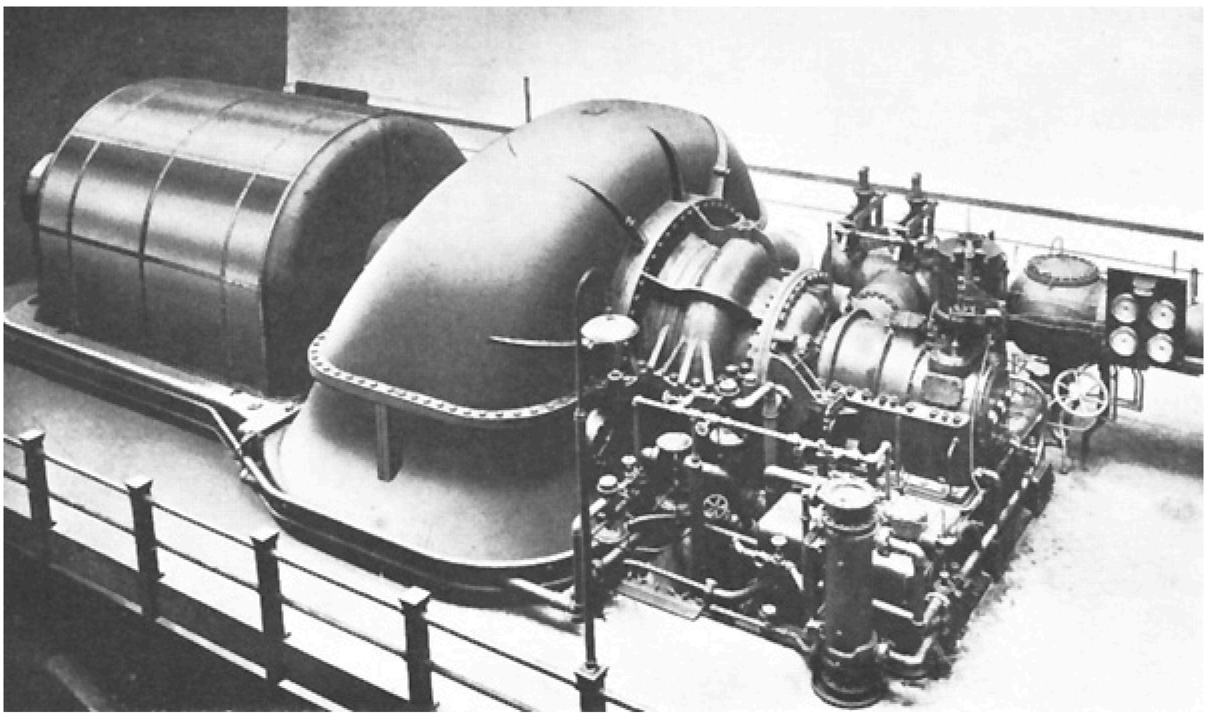


EXTERIOR OF MAIN POWER STATION AT SOUTH BOSTON

Briefly there are four main routes as follows:

Forest Hills-Everett (via tunnel)	8.59 miles
Forest Hills-Everett (via elev.)	9.35 miles
Harvard-Andrew	5.56 miles
No. Station West-Kenmore	2.87 miles
Bowdoin-Maverick Sq.	1.67 miles

The Forest Hills-Everett route is called the main line, and the Harvard-Andrew route the Cambridge Subway. The Bowdoin-Maverick Square line up to the present has been operating three-car trains with overhead trolley, but new equipment consisting of steel cars is now on order and the third rail is now being installed in the tunnel. The Lechmere Sq.-Broadway line over East Cambridge Viaduct and Tremont St. Subway is also considered a rapid transit route, although surface type cars are used with overhead trolley. These cars are equipped for multiple unit control and are operated in three-car trains.

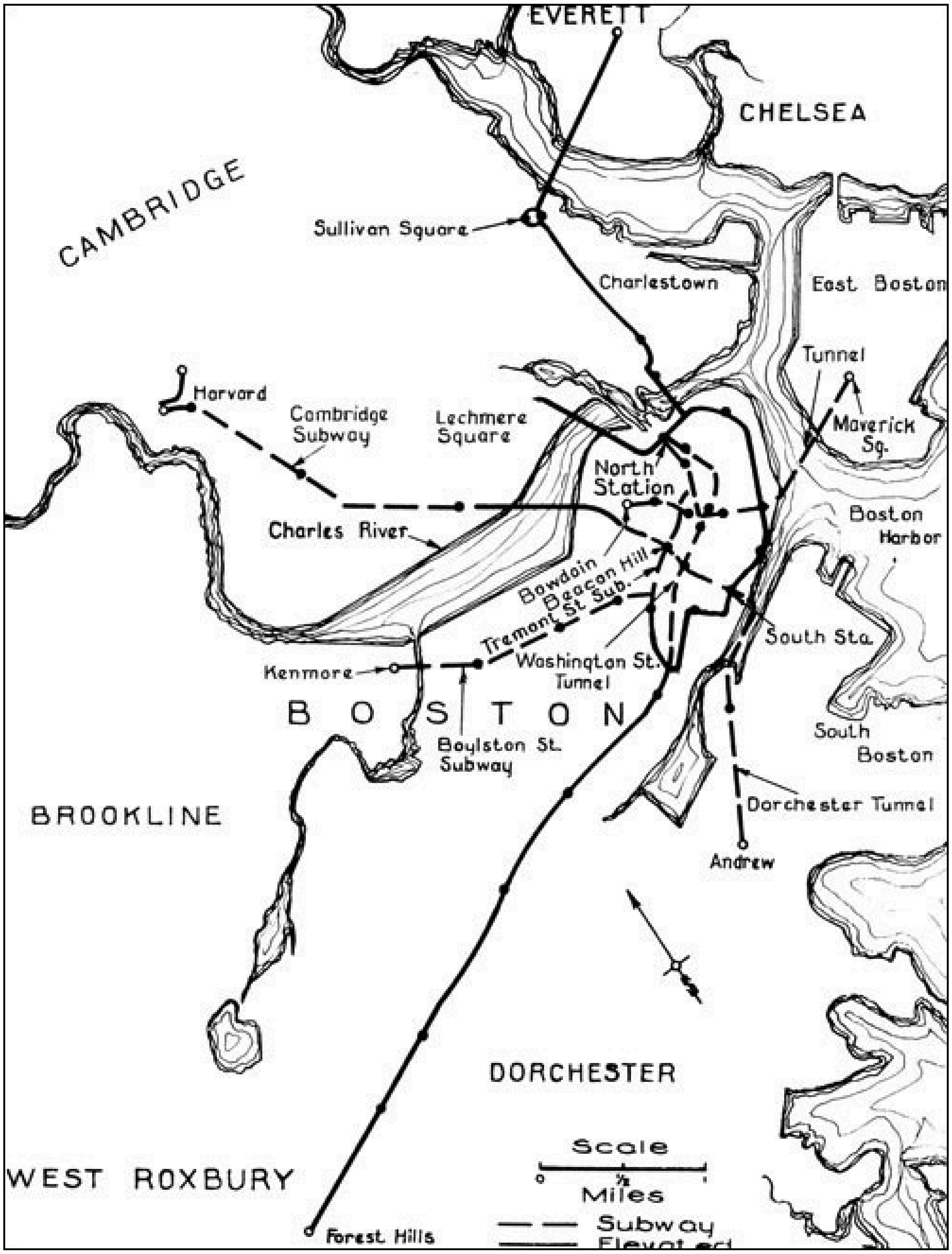


35,000-Kw. Turbo-Generator in South Boston Power Station

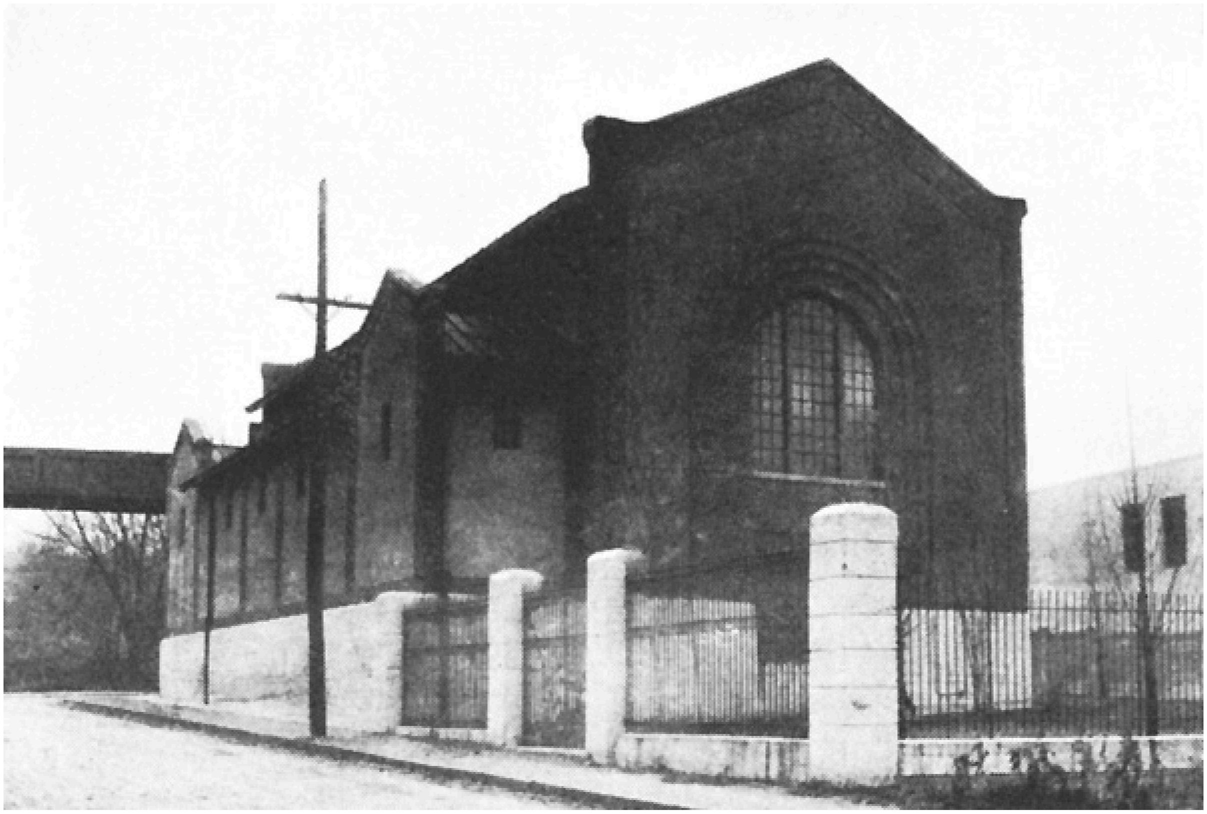
The rush hour trains on the main line include as high as eight cars, which is the limit set by the length of the station platforms. The signal system is entirely automatic and during rush hours the headway varies from 2 to 3½ minutes on the main line. The maximum grades encountered are 2 to 3 percent with a high percentage of heavy curvature. By taking advantage of the transfer arrangements at terminals, rides of 14 miles can be obtained for a single fare.

Power Station Equipment

The power system as originally installed included several engine-driven direct-current plants suitably located for distributing 600 volts direct to the trolley. With the extension of the system, however, an alternating-current station was installed at South Boston, generating 25-cycle three-phase current for distribution at 13,200 volts to synchronous converter substations. Alternating-current generating equipment has also been installed at the Lincoln Station. The total installed capacity of turbine stations is now 115,000 kw. while the direct-current generating stations have practically all been discontinued.



Rapid Transit Lines—Boston Elevated Railway



Exterior of Egleston Square Substation

Substations

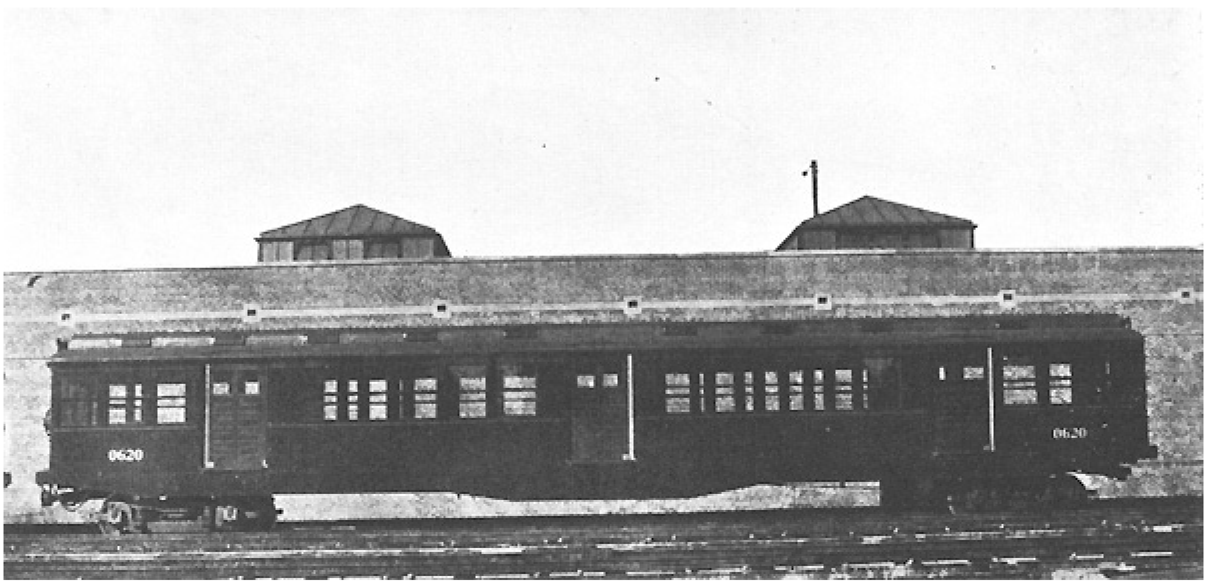
There are in operation for supplying power to both elevated and surface lines a total of 12 synchronous converter substations having a total rated capacity of 58,000 kw. The power consumption of the Rapid Transit lines is somewhat less than half the total energy used.

Distribution

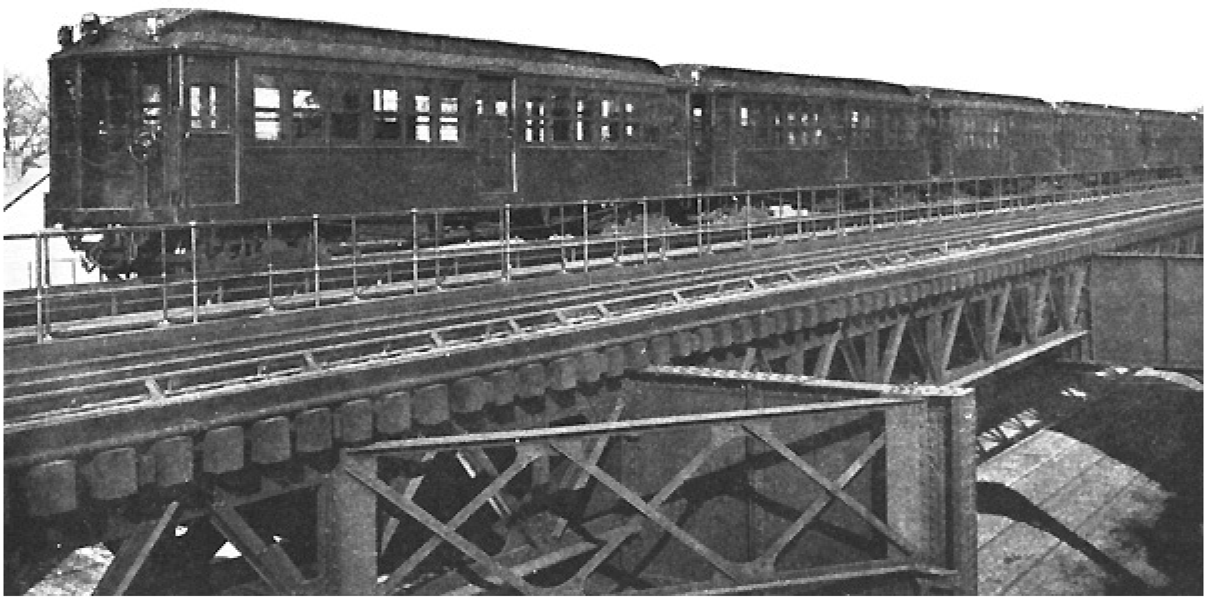
Direct current is distributed from the several substations at 600 volts and is collected on the rapid transit systems from an 85-lb. over-running third rail.

Rolling Stock

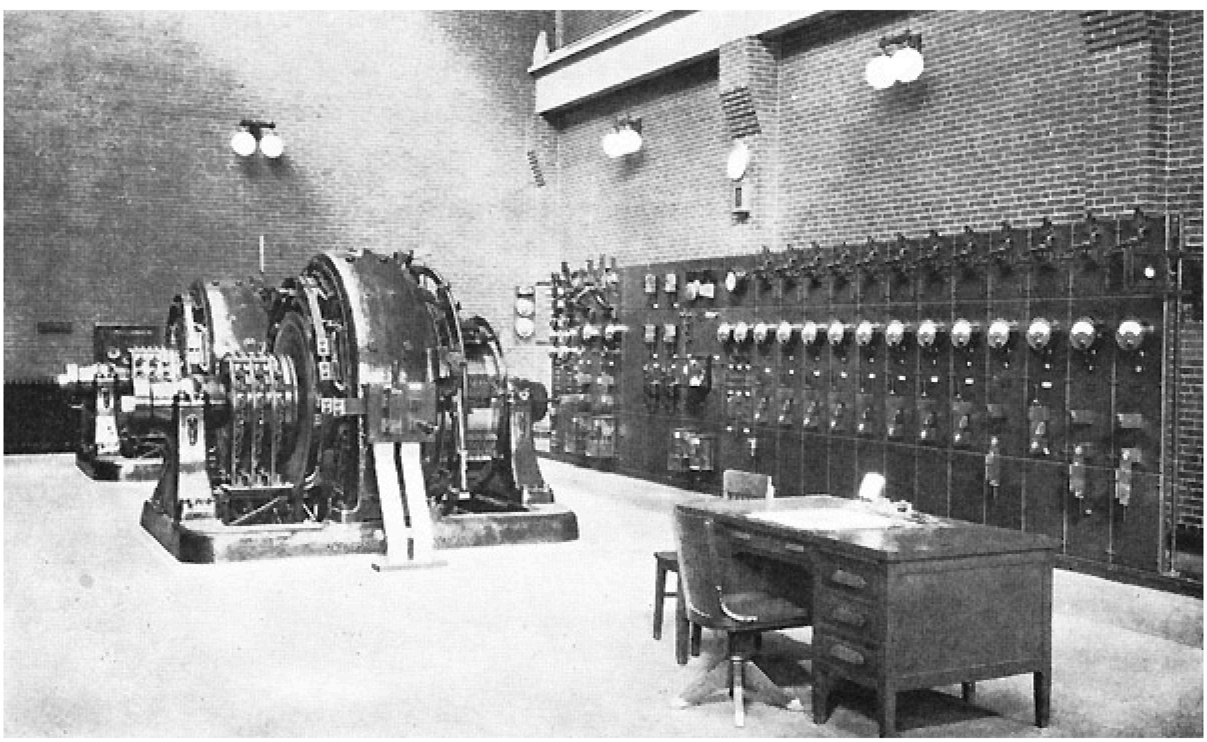
Altogether there are 420 cars in the rapid transit service, the older cars weighing about 34 tons with seating capacity of 48 and the newer type as used in the Cambridge Subway 43 tons each, arranged to seat 72 passengers. On account of the limiting clearances in the old subway the Cambridge cars cannot be used on the main line. All cars are motor cars and no attempt is made to use trailers. Each car is equipped with two motors and multiple unit control.



LATEST TYPE OF STEEL MOTOR CAR USED IN CAMBRIDGE SUBWAY



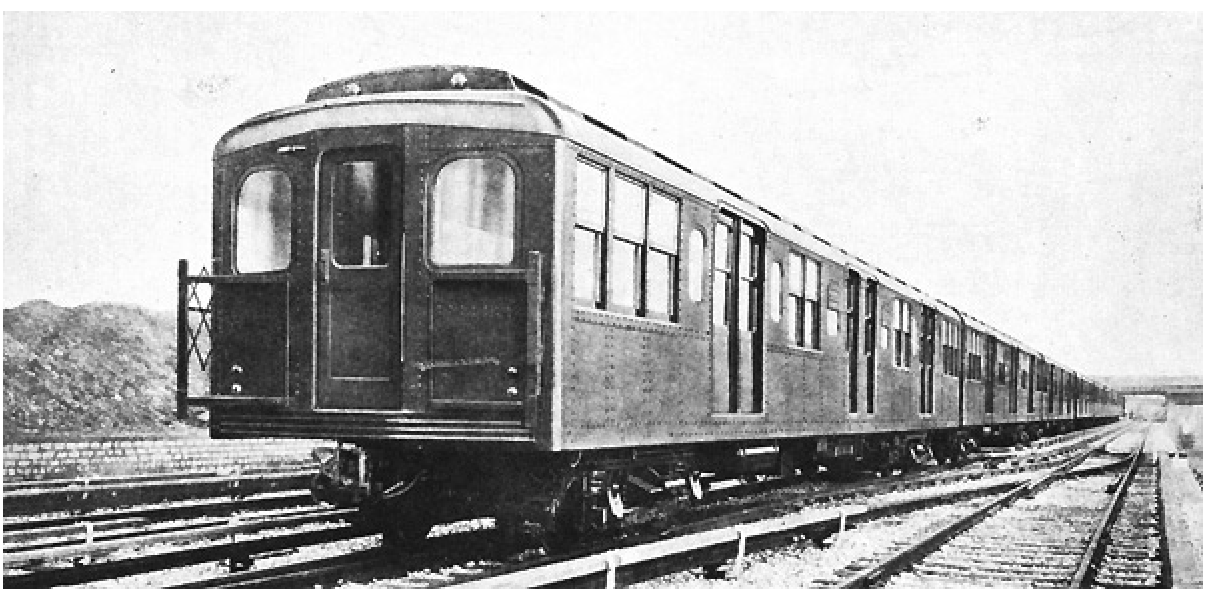
Main Line Train—Boston Elevated Railway



Interior of Substation Equipped with 2000-Kw. Synchronous Converters

BROOKLYN RAPID TRANSIT SYSTEM

The Brooklyn Rapid Transit Company controls all of the elevated and surface lines in Brooklyn including those reaching Coney Island. It also has entrance to Manhattan over the lines of the New York Municipal Railway Corporation, which was organized by the B. R. T. to finance and construct a part of the new city lines allotted to the B. R. T. The New York Municipal line runs through the new Broadway subway as far north as 60th St. thence east through the 60th St. tunnel under the East River to a connection with the Astoria and Corona lines in Queens. Other subway and bridge routes have been completed during the past few years as part of a definite plan, which contemplates the elimination of the present stub end operation at the lower end of Manhattan.



STANDARD NEW YORK MUNICIPAL MOTOR CAR EQUIPPED WITH GE-248 MOTORS

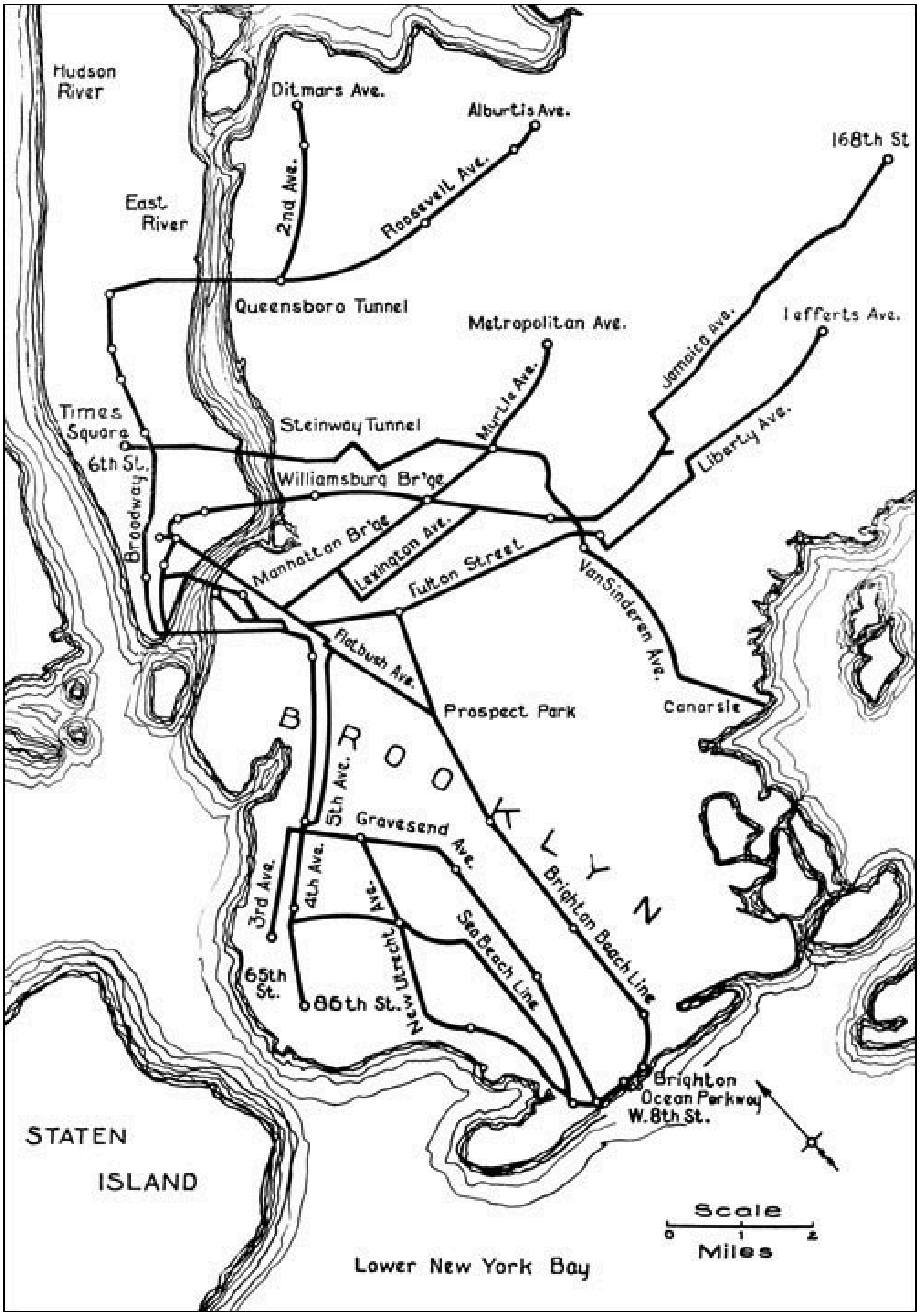
The Brooklyn Bridge line built in 1883 and the Brooklyn Elevated R. R. in 1888 formed the nucleus of the present Brooklyn Rapid Transit system. Electrical equipment was tried out in 1898 and additional motor cars were put in service in 1902. This improvement rapidly displaced the “steam dummies” and facilitated the extension of lines and the handling of a rapidly increasing traffic.

Of the present lines on the Brooklyn Rapid Transit system 89.20 miles of route aggregating 258.35 miles on a single track basis can be classed as rapid transit lines and operate multiple unit trains with third rail current collection. This includes the several elevated branches in Brooklyn and the newer subway lines of the dual system all of which are operated by the New York Consolidated R. R. Co., which is the operating organization.

The lines of the Brooklyn Rapid Transit system, which are operated by the New York Consolidated R. R., according to figures for the year ended June 30, 1921, handled 404,970,640 passengers over the rapid transit lines.

Power Supply

The original power equipment consisted of engine-driven direct-current generators, which have gradually been retired due to obsolescence.



RAPID TRANSIT LINES OPERATED BY NEW YORK CONSOLIDATED R. R. CO.

Power for operating the B. R. T. system is now generated in two alternating-current plants with installed capacities as follows:

Central (Third Av. & 2nd. St.)	16,500 kw.
Williamsburg (Kent Av. & Rush St.)	182,500 kw.

Power is generated and transmitted at 6600 volts, 25 cycles, three-phase. Owing to the diversified feeding system it is not possible to estimate the portion used by the elevated and surface lines. Power for the operation of the Manhattan lines is purchased from the Interborough Rapid Transit Co.

Substations

For supplying 600 volts to the entire system the B. R. T. has in operation 98 synchronous converter units aggregating 142,500 kw. These units range in size from 500 to 4000 kw. each. Many of the stations feed both elevated and surface lines so that it is difficult to approximate the capacity available for the rapid transit service.

Distribution

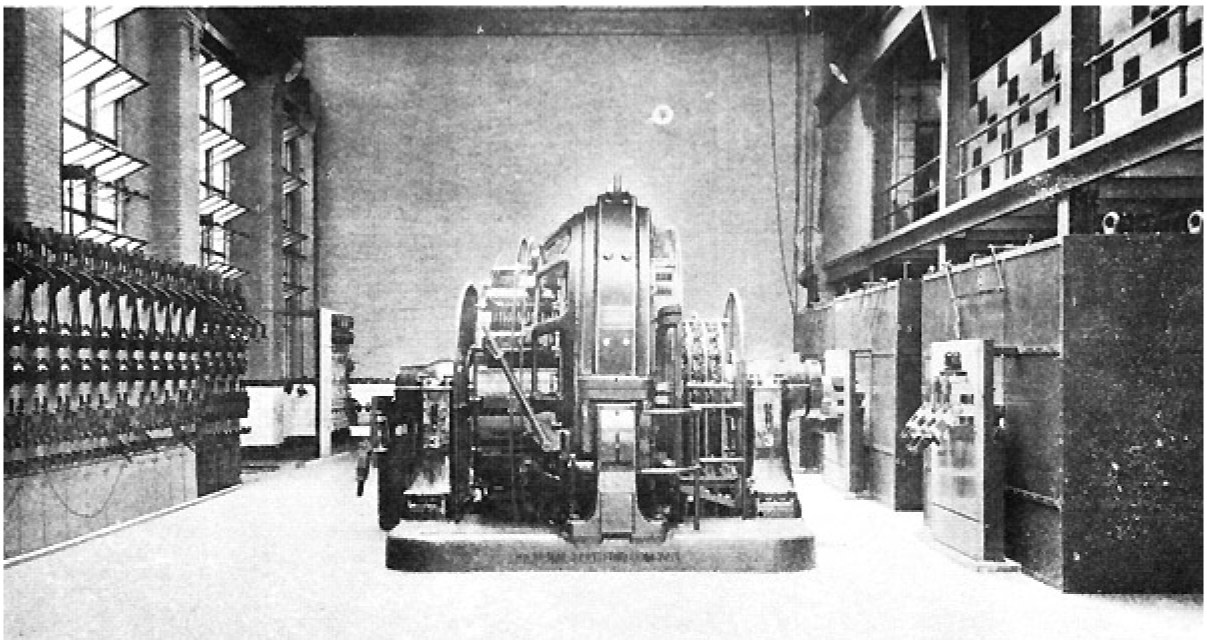
Current collection on all elevated and subway lines is from an over-running third rail. The following sizes of third rail are in use:

Early Elevated lines	55 lb.	(to be replaced with 80 lb.)
Subway lines	80 lb.	
New Subway	150 lb.	

Rolling Stock

The New York Consolidated R. R. Company operates in subway and elevated service a total of 1550 cars each equipped with two motors and multiple unit control. These include the equipment operated over the New York Municipal lines through the new subways. 900 of the newest cars use GE-248 motors and weigh, fully equipped, about 45 tons with seats for 72 passengers. These new cars are operated in all motor car trains.

Trains up to seven cars are operated in rush hour service and the minimum headways approximate two minutes. The maximum length of ride possible for a single fare is from Corona through the Broadway subway to Coney Island, about 21 miles. The maximum grade on the system is 5 per cent on the New York Municipal line.



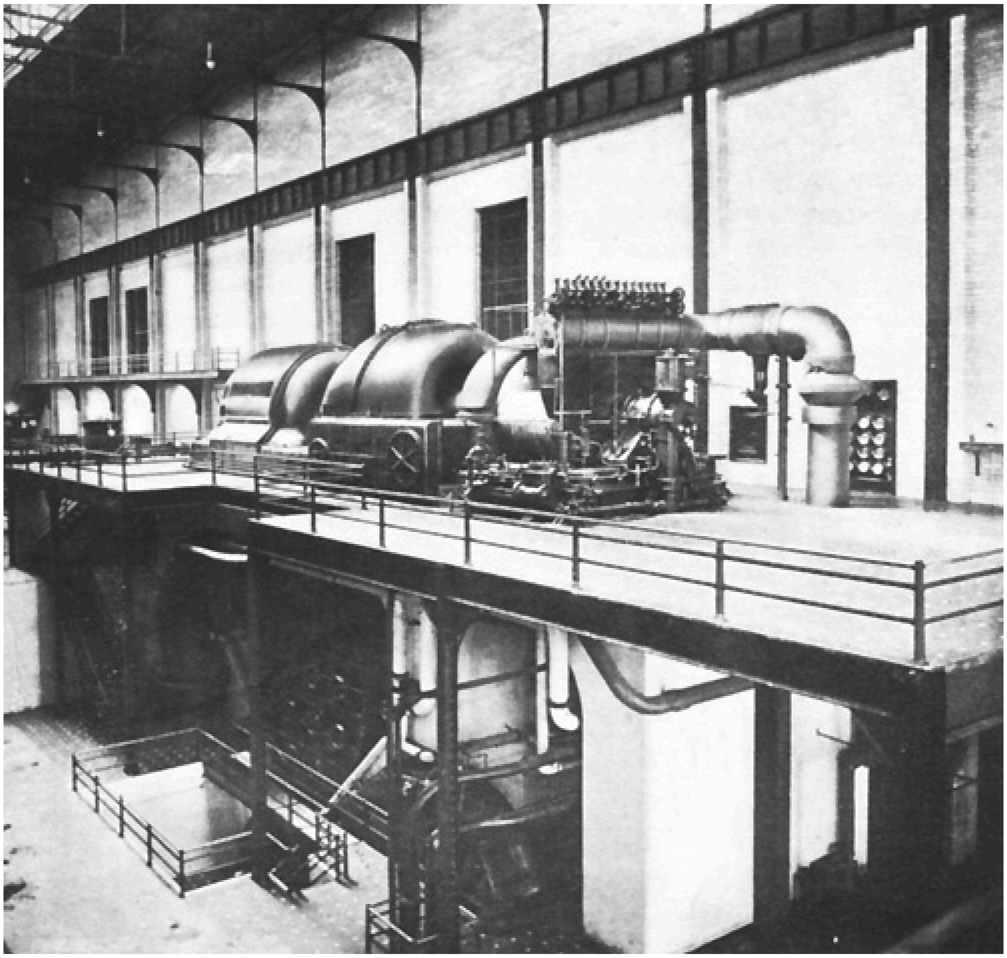
4000-Kw. Synchronous Converters Installed in South 6th Street Substation

CHICAGO ELEVATED RAILROADS

The present Chicago Elevated Railroads are an amalgamation of the four systems which up to 1911 were operated as independent lines. Under the unified system of operation a single fare takes the passenger from one end of the system to the other, except that north of Howard Street on the Evanston line an additional fare is collected. The longest continuous ride without change is from Wilmette to Jackson Park, a distance of 24 miles.

The first elevated road, afterward known as the South Side Elevated, started operation in June, 1892, with steam engines. After the successful demonstration on the Intramural Railway this line was electrified; all steam equipment being withdrawn in 1898.

What is now the Chicago and Oak Park Elevated Railroad began operation in 1893 also with steam locomotives. Electrical operation began in September, 1896.



30,000-Kw. Curtis Turbine in Northwest Station of Commonwealth Edison Company

The Metropolitan West Side was originally planned for steam locomotive operation, but developments in electric traction during the construction period were so rapid that orders for steam equipment were cancelled and operation began in May, 1895, with electric equipment.

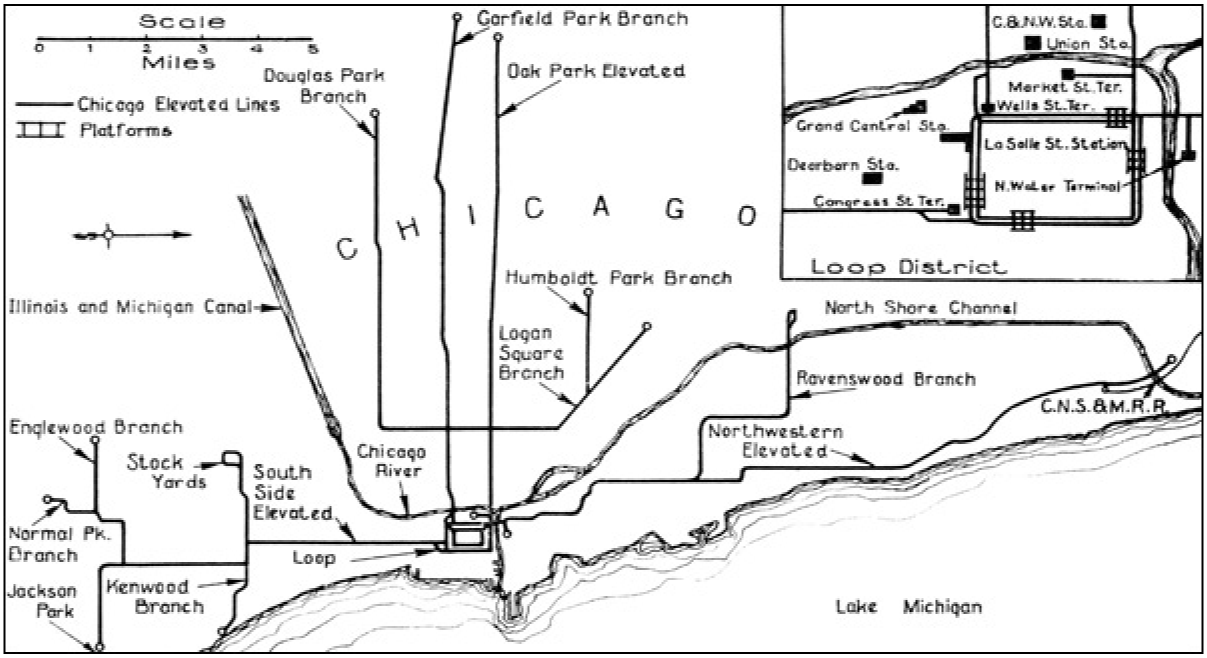
The Northwestern Elevated began operation in May, 1900, and was planned as an electric rad from the start. In 1897 the “Union Loop” was built to facilitate interchange of passengers from the different lines, but a separate fare was required on each road up to 1913.

The population served by the Chicago Elevated Lines is estimated at more than 1,000,000 people; the total number of passengers handled annually is about 190,000,000. Trains of from six to eight cars are operated during rush hour service on a two-minute headway with a maximum of 72 trains per hour on a track of the loop. Plans are being made to extend some of the station platforms to permit the use of more than 6- and 8-car trains.

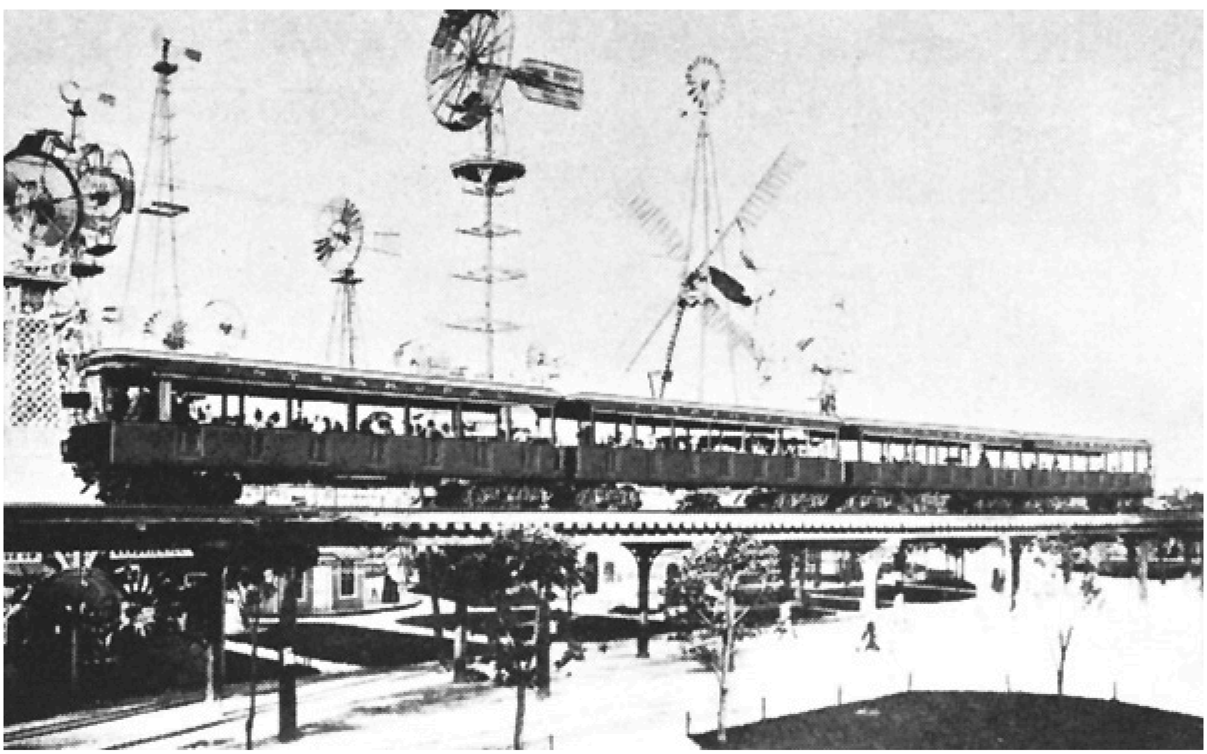
An extensive program of improvements to the present rapid transit system has been proposed, but no definite steps have yet been taken toward authorizing the work. These plans include a subway section under the present loop district with several additional elevated lines.

PRESENT MILEAGE OF CHICAGO ELEVATED LINES

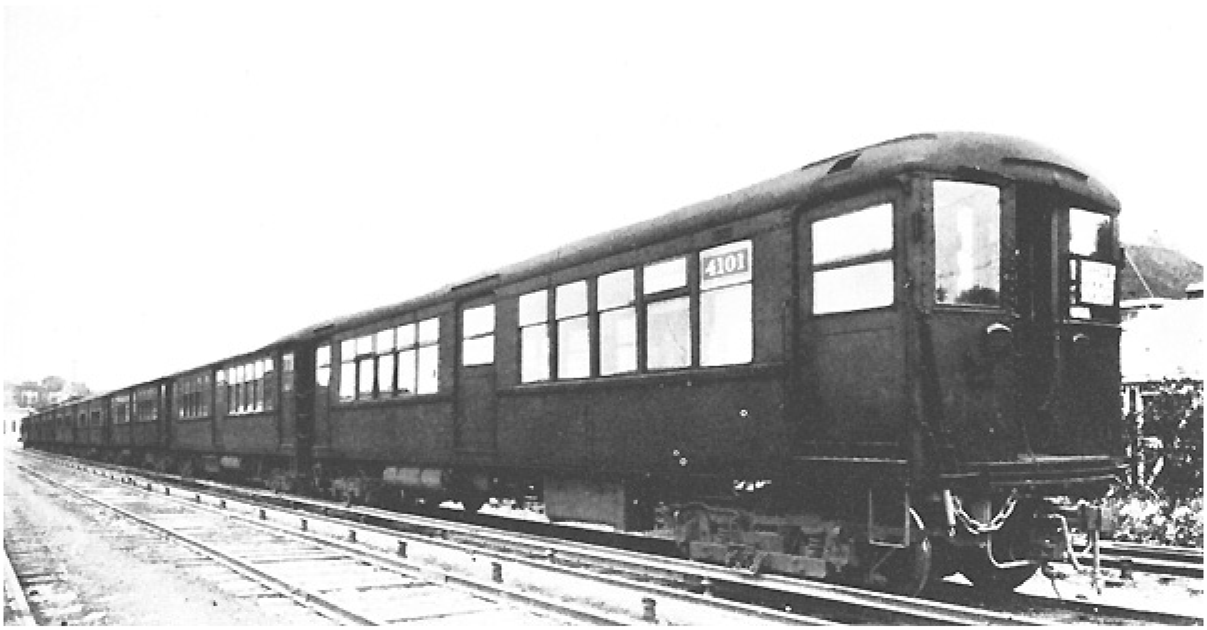
	Route Miles	**Single Track Miles**	**Yard**	**Total Track**
Northwestern Elevated	19.7	52.33	9.28	61.61
Chicago & Oak Park	9.32	20.38	2.28	22.66
Metropolitan West Side	23.83	53.63	7.78	61.41
South Side	16.15	35.99	9.97	45.96
Loop	2.12	4.72	....	4.72
	91.12	167.05	29.31	196.36



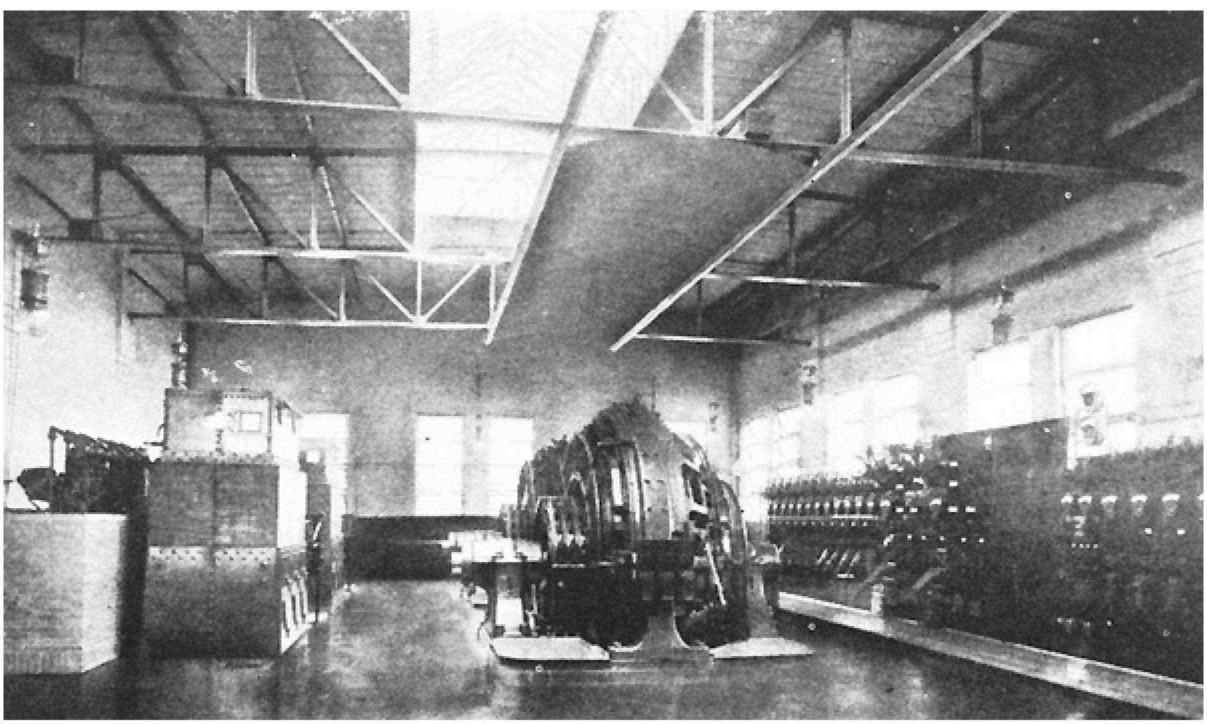
Chicago Elevated Railroads



Train on Intramural Railway Chicago, 1893



8-car Train—Chicago Elevated Roads—Equipped with GE-243 Motors



4000-Kw. Synchronous Converters at Campbell Avenue and Homer Street Station

Power Supply

The power for the early elevated lines was derived from engine-driven direct-current power plants all of which have since become obsolete. All power, therefore, is supplied from the network of the Commonwealth Edison Co., which maintains an ample reserve to supply all needs. While a large percentage of the power now being purchased is generated at 25 cycles, the policy of the Power Co. on all new equipment is to specify 60 cycles.

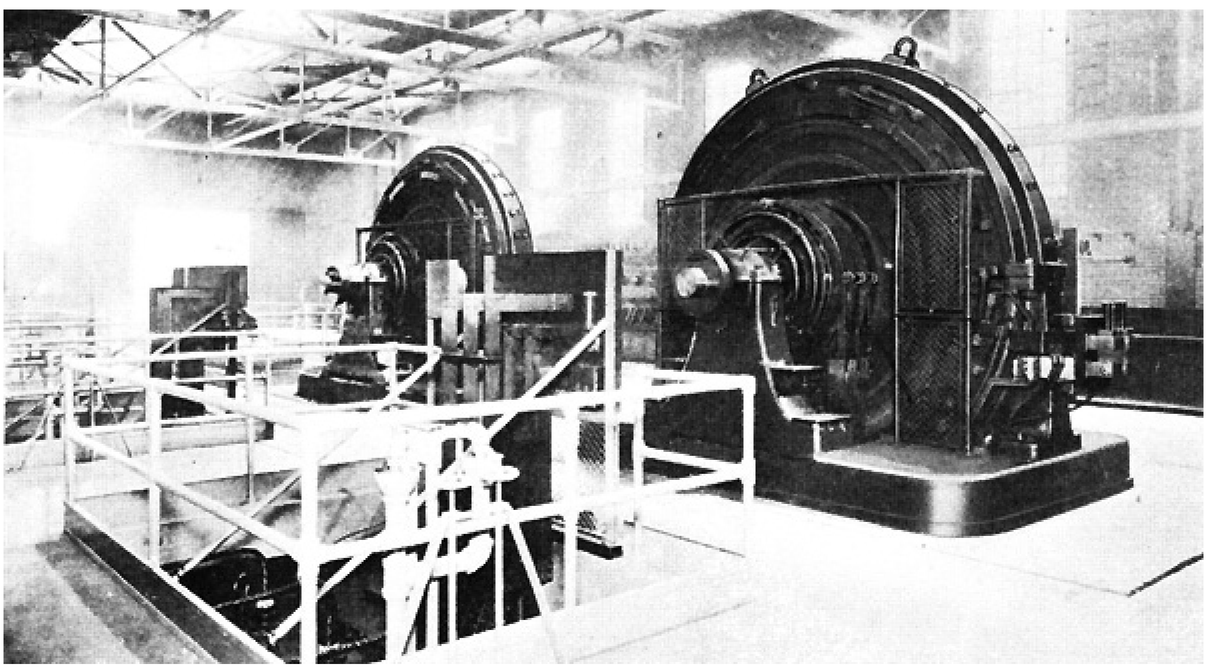
Substations

12 out of the 32 substations supplying the various traction systems are owned and operated by the Commonwealth Edison Co. and only 60-cycle generating equipment is installed when adding to their present capacity.

The several substations contain synchronous converter units ranging in size from 1000 to 4000 kw. each. The preferred size for new 60-cycle substations is the 3000-kw. unit of which there are now five in service. The following table shows the ownership and gross capacity of the substation equipment for all of the Chicago lines. It is not possible on account of the diversity of feeding arrangement to designate any particular stations as belonging exclusively to the elevated lines.

SUBSTATIONS—CHICAGO TRACTION SYSTEMS

Operating Co.	**No. Stations**	**No. Units**	**Total Capacity**
Chicago Railways Co.	10	32	80,000
Chicago City Railway	7	26	53,400
Calumet & So. Chicago Railway	3	9	9,000
Commonwealth Edison Co.	12	33	105,000
Elevated R. R.	3	8	9,000
Totals	35	108	256,400



Latest Type of 3000-Kw., 60-Cycle Synchronous Converters Installed by Commonwealth Edison Company, for Chicago Surface and Elevated Lines

The Northwestern Elevated R. R. has on order a complete 2000-kw. automatic substation from the General Electric Co. to be installed at Buena

Park. This is the first application of the automatic to Metropolitan Rapid Transit service.

Distribution

Energy for elevated train operation is fed to the third-rail shoes at 600 volts. The third rail is of the top contact unprotected A.S.C.E. rail varying in size from 40 to 80 lb.

Rolling Stock

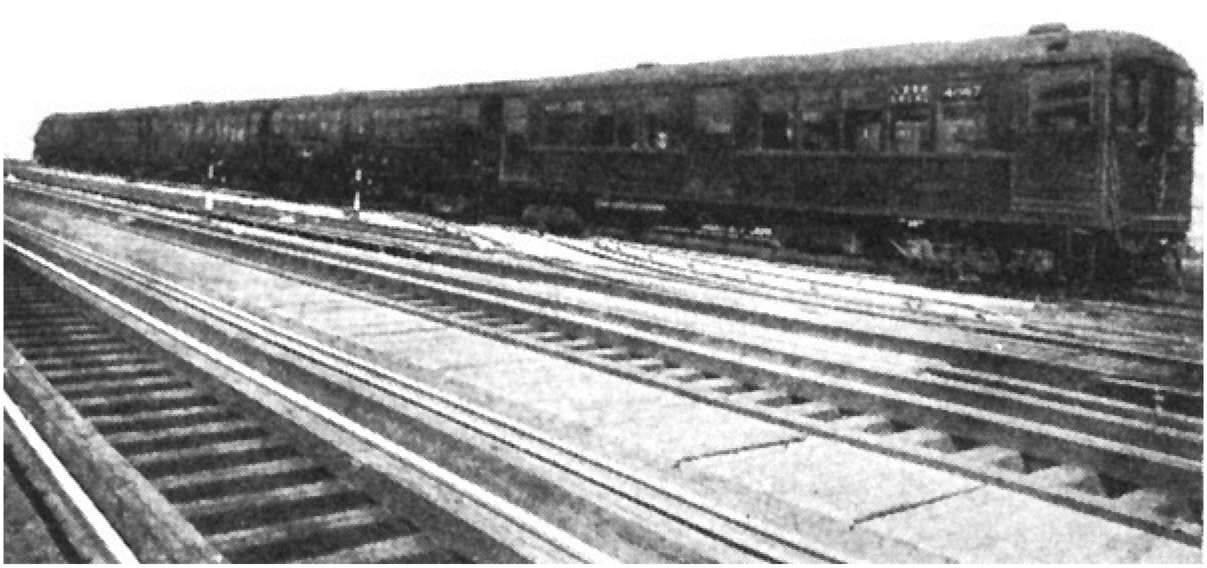
The rolling stock equipment includes a total of 1008 two-motor cars weighing from 22 to 35 tons each, the latter figure representing the more recent types of cars. The distribution of these cars among the four divisions is as follows:

	No. Motor Cars
Northwestern Elevated	302
Chicago & Oak Park Elev.	84
Metropolitan West Side Elev.	253
South Side Elev.	369
Total	1,008

In addition to the motor cars there are available for use on the several divisions about 660 coaches which can be used as trailers.

Supplemental to the regular elevated service the Chicago Elevated System affords entrance to the business section of the city to the Chicago, North Shore & Milwaukee R. R. an affiliated line operating a high speed interurban service between Chicago and Milwaukee. This line enters from the north operating over the Northwestern division at Evanston.

Connection is also made at Des Plaines at the end of the Garfield Park Branch with the Chicago, Aurora & Elgin R. R., a high-speed third-rail line reaching Aurora, Elgin, Batavia, Geneva and other points west. These trains also enter the city running over the Metropolitan West Side tracks into the loop district.



4-CAR TRAIN ON NORTHWESTERN ELEVATED EQUIPPED WITH GE-243 MOTORS

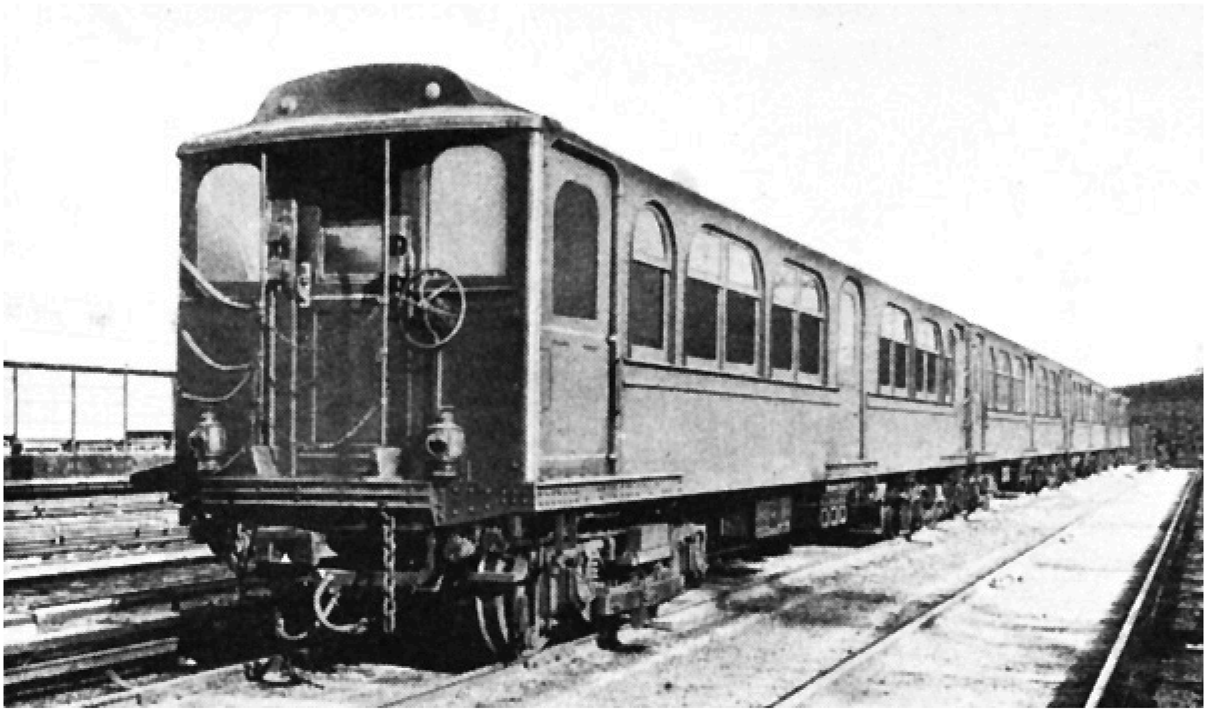
HUDSON & MANHATTAN R. R.

The Hudson & Manhattan R. R. Conducts a Heavy Passenger Traffic Between Lower Manhattan and Jersey City points and between an uptown station at 33rd. St. and Hoboken, N. J. Rapid Transit service is also maintained between Hudson Tunnel and Newark over the tracks of the Pennsylvania R. R. These lines popularly known as the Hudson Tubes are to a large extent operated through tunnels under the Hudson River. Direct under-river connection is made between New York and the stations of the Erie, D. L. & W. and Pennsylvania Railroads.

The total mileage of the system is made up as follows:

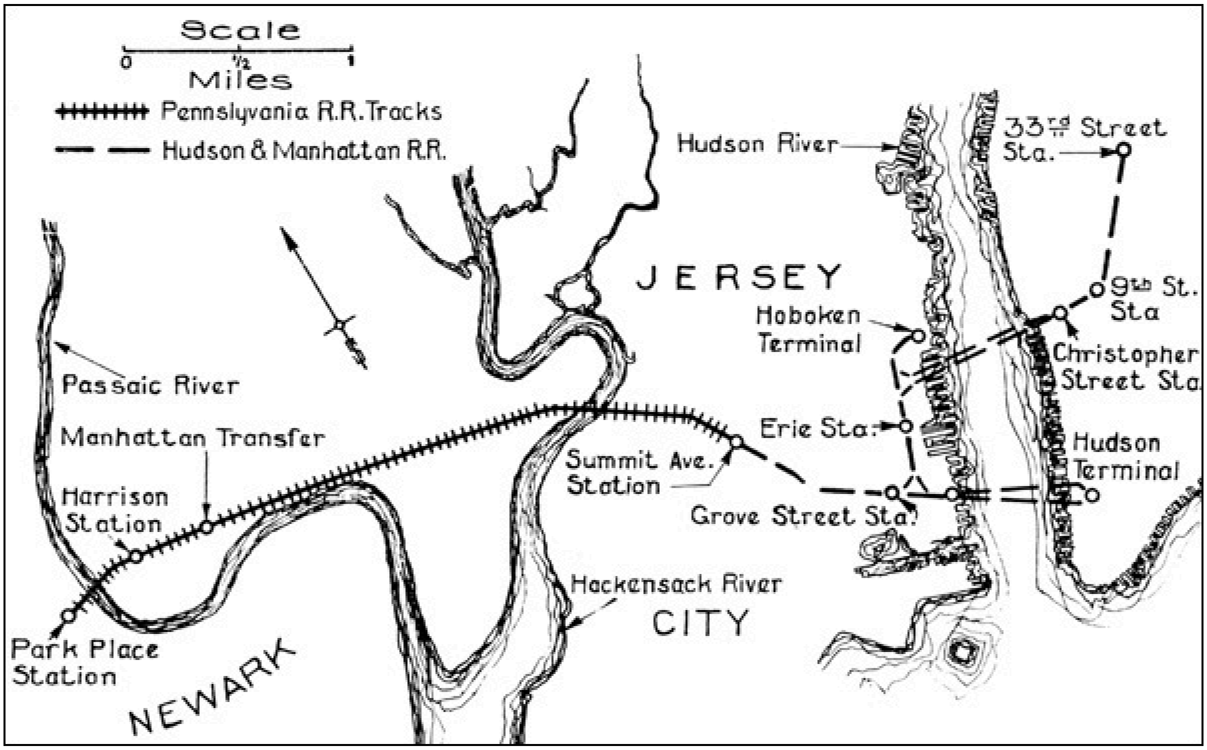
Miles of road	7.869
Extra track	8.634
Sidings, etc.	.332
Car Houses and Shops	1.920
Total	18.768

The road was opened in February, 1908, for transportation of passengers from Jersey City to lower Manhattan and later to the uptown terminal.

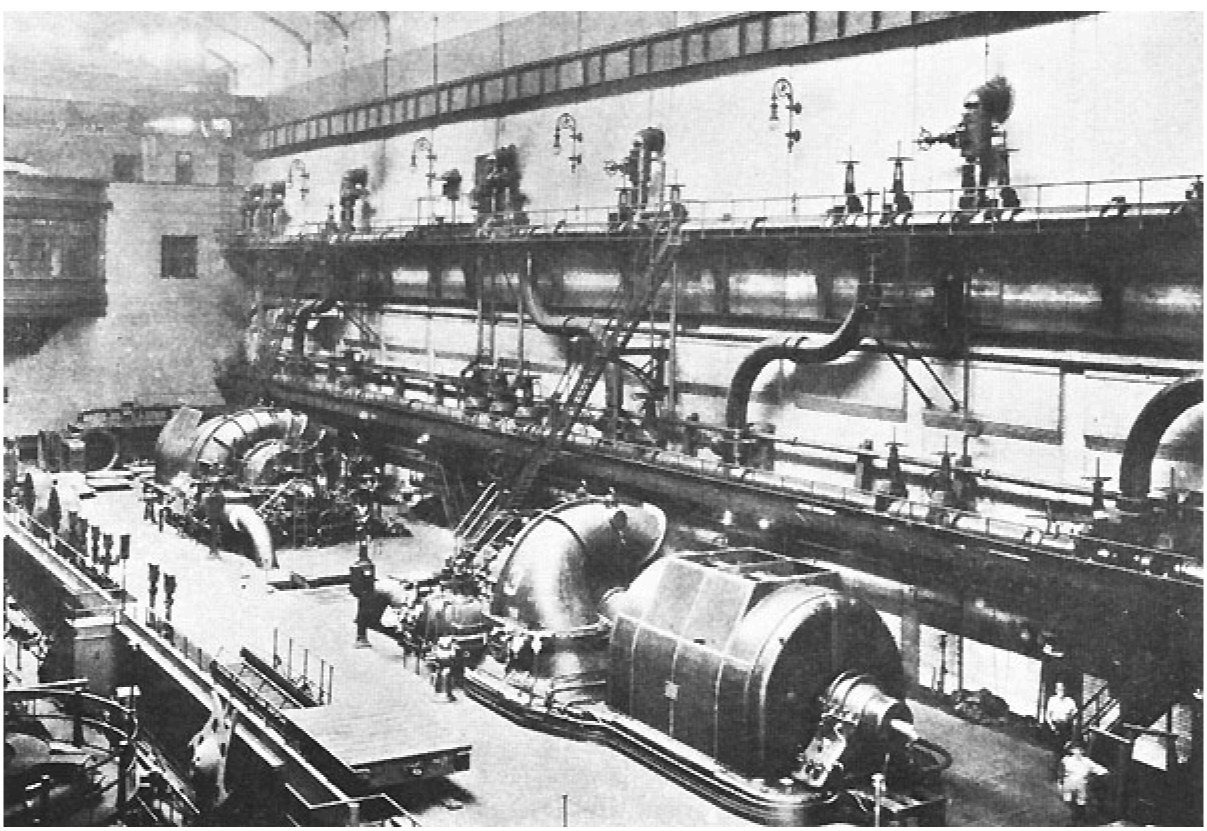


6-car Train on Hudson & Manhattan Railroad Equipped with GE-76 Motors and Type M Control

3-car Train for Newark Rapid Transit Service Equipped with GE-212 Motors and Type M Control



HUDSON AND MANHATTAN RAILROAD



Two 35,000-Kw. Curtis Turbines in Waterside Station No. 1 New York Edison Company

The electrical equipment, which was furnished throughout by the General Electric Company, includes:

A power station equipped with four Curtis turbo-generating units aggregating 18,000-kw.

Three substations containing a total of 11-1500-kw. synchronous converters.

311 electric motor cars equipped for multiple unit operation. (60 of these cars operated in the Newark Suburban service are the property of the Pennsylvania R. R.)

POWER SUPPLY

The power station was equipped with two 6000-kw. and two 3000-kw. vertical Curtis turbines generating 25-cycle three-phase alternating current at 11,000 volts. This plant is at Jersey City conveniently located for the reception of coal for fuel and use of Hudson River water for condensing purposes. Through an arrangement agreed to some time ago power is now being purchased from the New York Edison Company, who have furthermore taken over the power station.

Substation No. 1 is located at Christopher & Greenwich Sts.; No. 2 at Washington & First Sts. (in Power House) and No. 3 in the Hudson Terminal Building. The 600-volt current for train propulsion is distributed through a 75-lb. top contact third rail reinforced with suitable feeders.

ROLLING STOCK

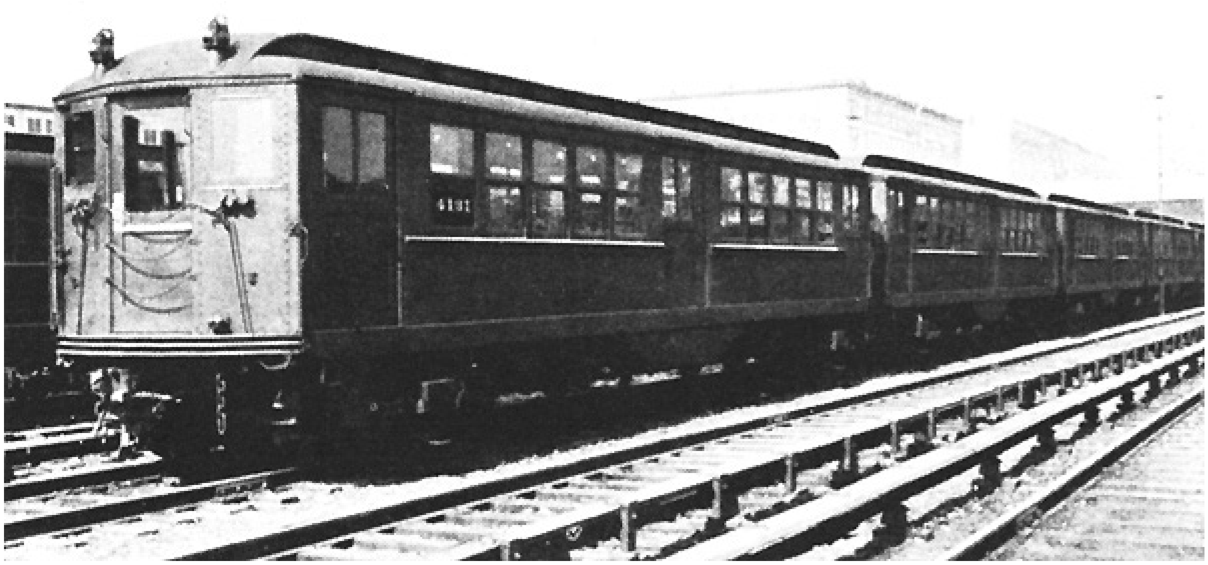
The motor car equipment owned by the Hudson & Manhattan R. R. includes 251 units, all motor cars, each carrying two motors either GE-76, GE-212 or GE-259 and Type M control. The cars are of all-steel construction and weigh from 35 to 37 tons each without passenger load. The Newark service requires 96 cars of which 36 are owned and 60 are furnished by the Pennsylvania R. R. Co. All of these cars are equipped with GE-212 motors.

Train service is operated on a uniform headway varying the number of cars per train to suit the traffic. Platforms are 370 ft. in length, which is sufficient to accommodate 8-car trains. The total number of passengers handled during the year ended June 30, 1921, was 95,607,645.

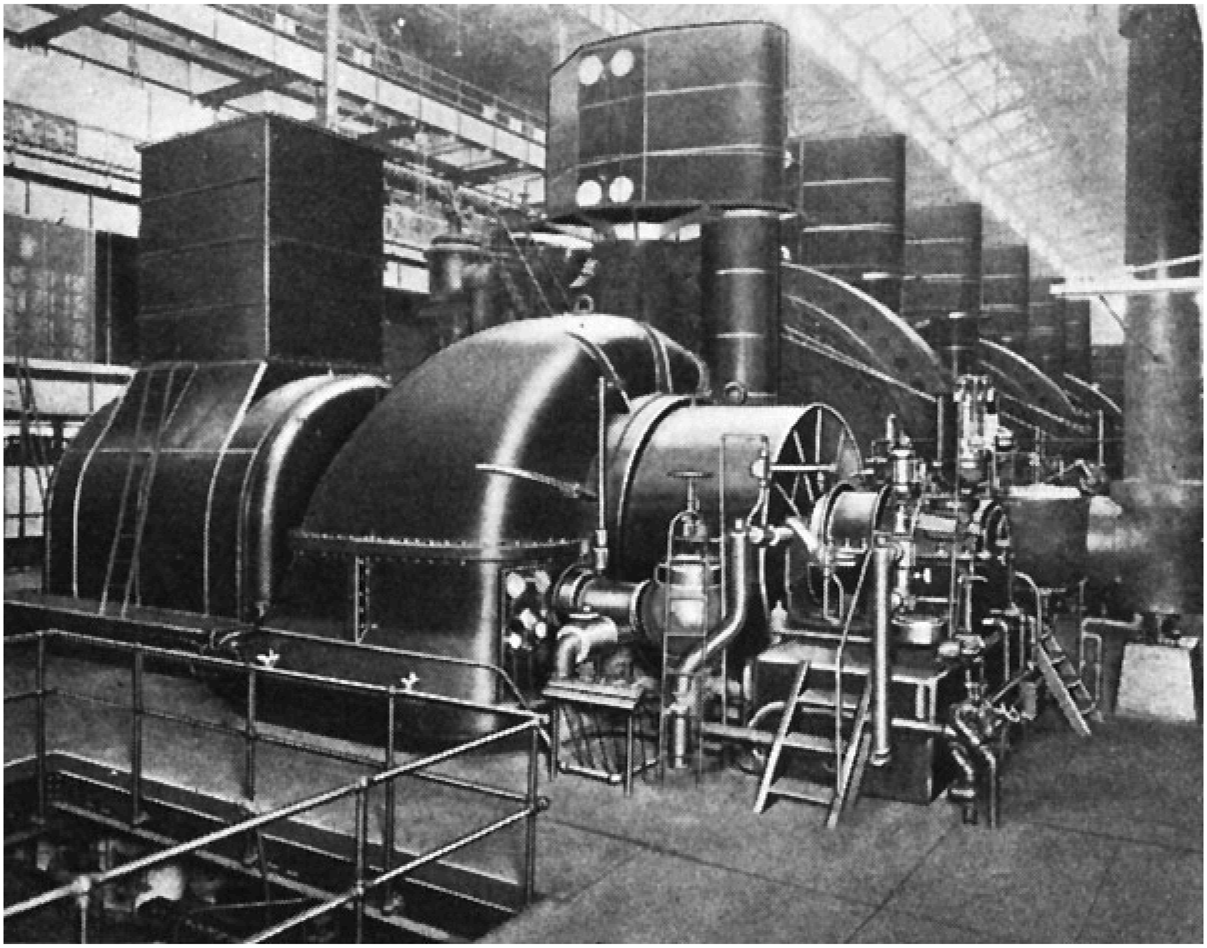
INTERBOROUGH RAPID TRANSIT CO.

On account of its geographical peculiarities the city of New York has for many years been subject to traffic congestion on the north-south line. The long narrow outline of the island of Manhattan with its dense population presents an unusually difficult transportation problem.

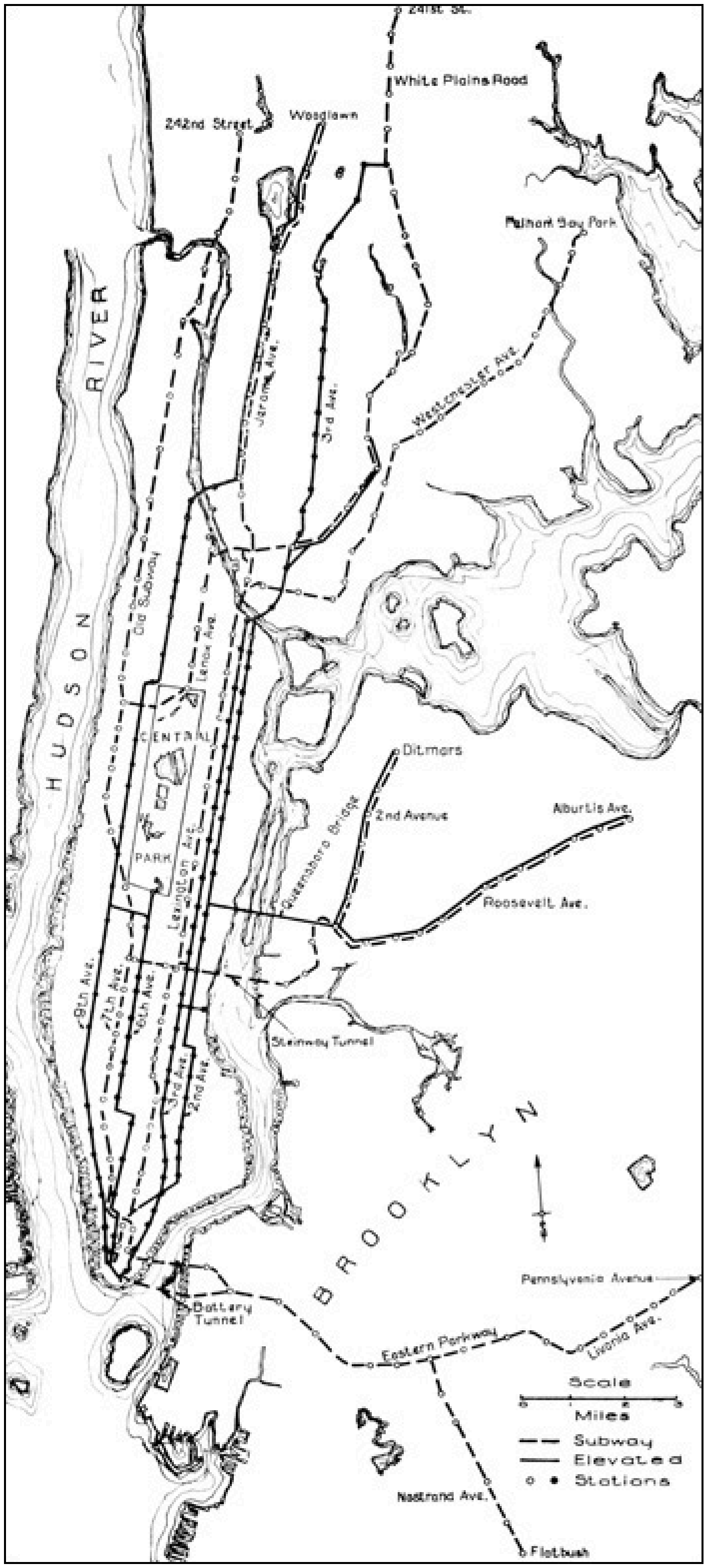
Until the year 1914 the operations of the Interborough Rapid Transit System were confined almost exclusively to Manhattan and the Bronx, while the Brooklyn Rapid Transit System operated in Brooklyn and the Borough of Queens. This geographical division, however, was abandoned with the inauguration of the dual system. By the new arrangement the B. R. T. operates into Manhattan over the New York Municipal line and on the other hand the Interborough reaches into Queens through the Steinway Tunnels and the Queensborough Bridge and into Brooklyn by the Eastern Parkway and Nostrand Ave. Lines.



7-CAR TRAIN EQUIPPED WITH G-E MOTORS AND PC CONTROL



30,000-KW. TURBINE AT 59TH STREET STATION, INTERBOROUGH RAPID TRANSIT



INTERBOROUGH RAPID TRANSIT LINES

The Interborough operates the old subway traversing the length of Manhattan and also the four elevated lines in Manhattan and the Bronx. The original lines of this system were built in 1868 and were the first elevated tracks in New York and probably the first in the world. Other elevated lines were built between 1870 and 1880, and the present old subway was completed in 1904-8.

Steam locomotives were superseded on the Manhattan Elevated line in 1902 and electric motor car trains were substituted. The original General Electric equipment installed at that time is still in daily operation after 20 years' service.

The present rapid transit facilities of the Interborough include the following branches:

Division	Length of Road	Total Single Track
Bronx Subway Line Contracts 1 & 2		
Underground	19.56	62.97
Elevated	6.16	22.32
Contract No. 3		
Atlantic Ave.	....	.27
Astoria Line Elevated	2.33	6.87
Brooklyn Line	5.10	17.66
Corona Line Elevated	4.21	12.61
Clark St. Tunnel	2.31	4.67
Jerome Ave. Line	6.04	18.89
Lexington Ave. Line	5.00	21.15
149th St. Loop	.55	.55
Nostrand Ave. Branch	2.70	5.55
Pelham Bay Park Line	7.15	21.6
Queensboro Subway		
Underground	1.61	5.98
Elevated	1.03	....
South Ave. Line Underground	4.19	15.73

White Plains Rd. Line Elevated	4.88	15.82
Manhattan Division Elevated	37.67	113.19
8th Ave. & 162nd. St. Connection	.62	1.26
Queensboro Bridge Line	1.35	2.73
Webster Ave. Line	1.74	5.33
West Farms Subway Connection	.5	1.00
Totals	114.7	373.15

The number of passengers carried by the Interboro Lines during the fiscal year 1921 was 1,013,678,831. This figure represents 2,773,479 passengers per mile of track. In the main 4-track subway 10-car express trains are operated during rush hour periods on minimum headways of 108 seconds.

Power Supply

The power generating equipment of the Interborough includes briefly the following:

59th St. Power Station		**Total Kw.**	
Turbo-Generators	3-30,000 kw.	90,000	
Compound Units	5-15,000 kw.	75,000	
Engine-driven Units	4-7,500 kw.	30,000	
Total		195,000	Kw.

74th St. Power Station		**Total Kw.**	
Turbo-Generator	1-60,000 kw.	60,000	
Turbo-Generators	3-30,000 kw.	90,000	
Turbo-Generator	1-7,500 kw.	7,500	
Engine-driven	3-7,500 kw.	22,500	
Total		180,000	Kw.

Power is generated 11,000 volts three-phase 25 cycles and transmitted principally underground at 11,000 and 19,000 volts. The total energy generated in the two main sections at 59th St. and 74th St. for the year 1921 was 830,000,000 kw-hrs.

Substations

For supplying 625-volt direct current to the rapid transit lines, there are a total of 25 substations containing 129 synchronous converters aggregating 283,000 kw.

Distribution

Propulsion current is delivered to trains through an unprotected over-running third rail weighing, in the old subway 75 lbs. per yard, on the elevated 100 lbs., and in the new subway 150 lbs. per yard.

Rolling Stock

The motor car equipment on the Manhattan Elevated lines includes over 800 cars which have been in operation since 1902-4 with GE-66 motors and Type M control. These cars after 20 years of hard service are referred to as the "back-bone of the system." Frequent additions have been made to elevated and subway equipment so that the total rolling stock at the end of the fiscal year 1921 was as follows:

INTERBOROUGH RAPID TRANSIT ROLLING STOCK EQUIPMENT

MANHATTAN DIVISION	
Passenger Motor Cars	1016
Passenger Trailers	721
Service Motor Cars	4
Service Trailers	56
ELEVATED EXTENSIONS	
Passenger Motor Cars	476

SUBWAY DIVISION CONTRACTS 1 & 2	
Passenger Motor Cars	785
Passenger Trailers	352
Service Motor Cars	10
Service Trailers	46
SUBWAY DIVISION CONTRACT 3	
Passenger Motor Cars	581
Passenger Trailers	217
Service Motor Cars	1
Total Motor Cars	2873
Total Trailers	1392
Grand Total	4265

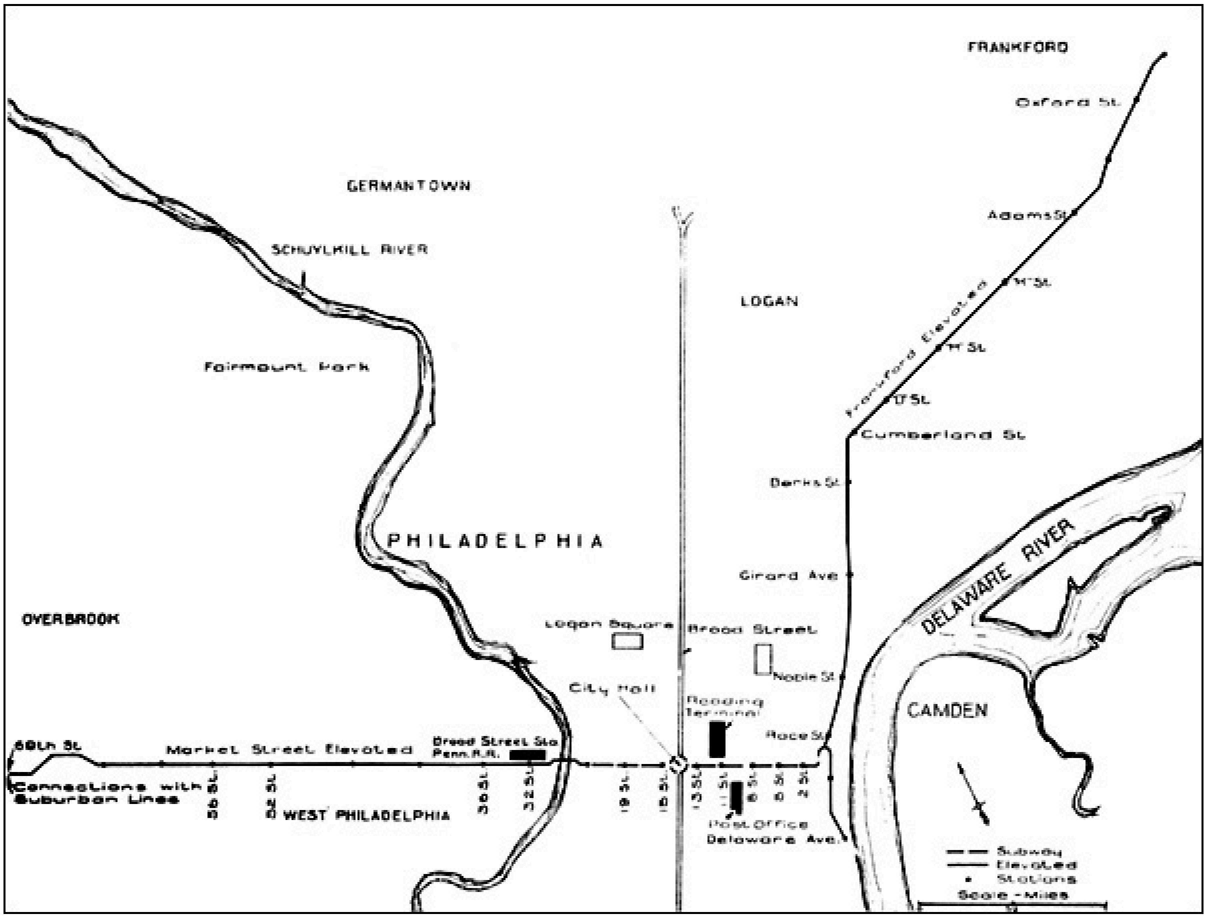
The longest ride on the system for a single fare is from the terminus of the White Plains Road line at 241st St. near the Northern boundary of the city, the entire length of Manhattan and the Bronx and through the Eastern Parkway Subway to Linwood Avenue, a distance of about 26 miles. The maximum grades encountered are 4½ per cent.

PHILADELPHIA RAPID TRANSIT CO.

The present rapid transit facilities of the City of Philadelphia include Market St. Subway-Elevated line extending East from the 69th St. Terminal to the Delaware River. The tracks are elevated from 66th St. to 22nd St. and pass in subway under the business section to another elevated section on Delaware Ave. This line first began service in 1905 and during the year 1920 handled approximately 80,000,000 passengers.

As far back as 1912 an exhaustive study of the city’s transportation facilities was made and a comprehensive program of extensions was proposed for the rapid transit system. Owing to legislative delays, and conditions due to the war, progress has been delayed on this program so that so far only the Frankford Elevated line has been built. This is now nearly ready for operation, from the foot of Market Street to Frankford, a distance of 6.4 miles all double tracked. Other extensions planned for construction in the near future include a four-track subway running north and south under Broad Street, and an elevated line extending from the present Market Street line at West Philadelphia to Darby.

The present elevated-subway system is double tracked throughout and multiple unit trains up to seven cars each are operated on headways down to two minutes. No express service is operated, all trains making every stop.



PHILADELPHIA RAPID TRANSIT ELEVATED AND SUBWAY LINES

TRAIN ON MARKET STREET ELEVATED, PHILADELPHIA RAPID TRANSIT COMPANY

Power Supply

The Philadelphia Rapid Transit Company's principal power station is at Delaware Avenue. Steam turbine generating equipment totalling 58,000 kw. is in service in three stations and is designed for 13,200 volts 3-phase 25 cycles at which it is transmitted to the substations. One or two direct-current stations are still available for supplying 600 volts direct to the trolley. Power is also purchased from the Philadelphia Electric Co. and the Philadelphia Hydro-Electric Co.

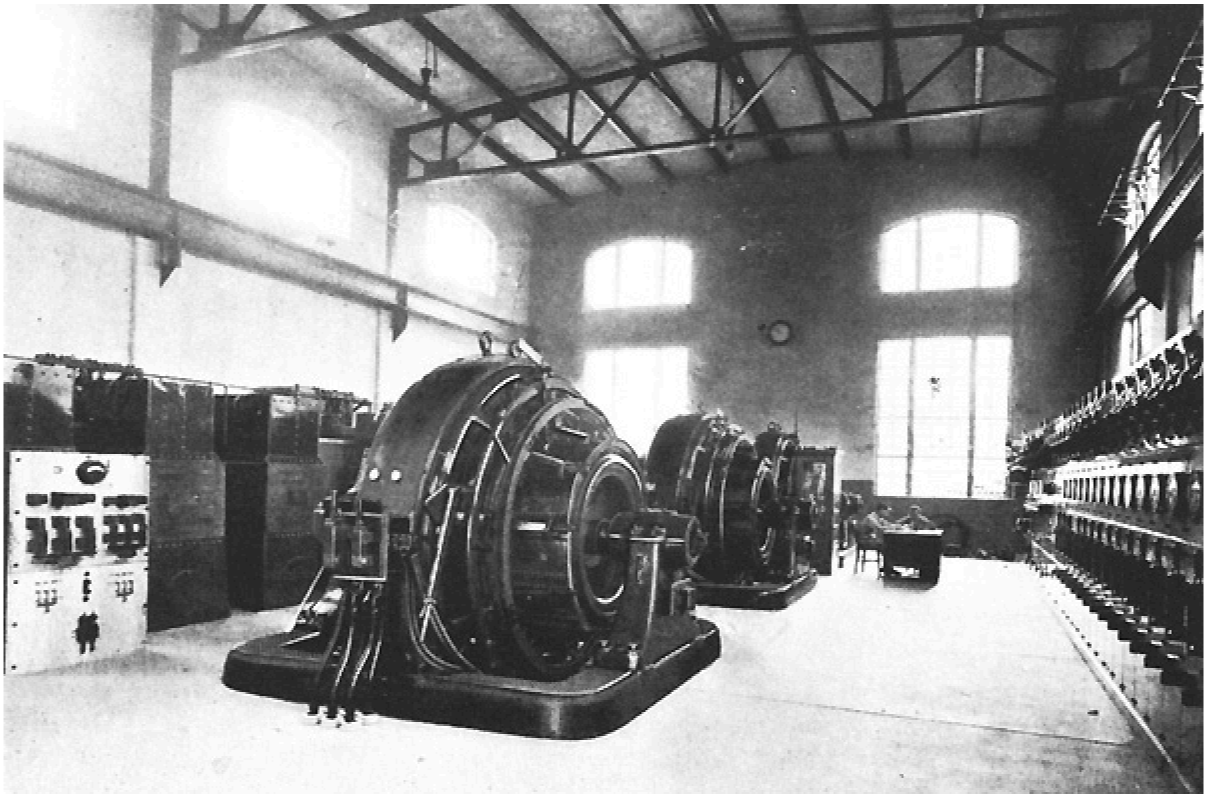
Substations

The company operates a total of 17 substations used for supplying both surface and rapid transit lines. These stations contain a total of 65 units aggregating 103,500 kw.

Power is distributed to all lines at 600 volts and on the rapid transit lines is collected from an under running third rail similar to that used on the New York Central Electric Zone.

Rolling Stock

The rolling stock used on the Elevated-Subway line includes 215 motor cars each equipped with two motors. Trains are made up of all motor cars, no trailers being used. General Electric motors are used throughout including GE-66 and GE-222. The Frankford extension will be operated with 100 motor cars each equipped with two GE-259 motors.



INTERIOR OF SUBSTATION AT 15TH AND TUCKER STREETS, SHOWING 2000-KW. SYNCHRONOUS CONVERTERS

GENERAL ELECTRIC COMPANY EQUIPMENT IN MULTIPLE-UNIT
SUBWAY & ELEVATED SERVICE

SYSTEM	**Cars**				**Motors**		
	No.	Yr. put in Service	Total Wgt. Pounds	Length Overall Ft. In.	No.	Type	Trailer Operating
Boston Elev. Rwy. Co.	40	1917	66383	46 7¼	2	GE-259-B	}
	45	1920	66628	46 7¼	2	GE-259-B	} No
	24	1920	68009	46 7¼	2	GE-259-B	}
	20	1912-3	86400	69 6½	2	GE-212	}
Hudson & Manhattan R. R.	50	1907	74550	48 3	2	GE-76	}
	90	1909	69620	48 5	2	GE-76	}
	50	1910	69620	48 5	2	GE-76	} No
	96	1911	73000	48 3½	2	GE-212-C	}
	25	1921	73500	51 3½	2	GE-259	}

Interborough Rapid Transit Co.	828	1902-3	75500	47 0½	2	GE-66-A	}
	200	1904	58500	47 0½	2	GE-69-B	}
	91	1907	59160	47 0½	2	GE-211-A	}
	40	1907	83200	47 0½	2	GE-212-A	}
	190	1909	83200	51 0½	2	GE-212-A	} Yes
	6	1915	70960	51 0½	2	GE-240-C	}
	161	1915	75000	51 0½	2	GE-259-A	}
	103	1915	75500	51 0½	2	GE-260-A	}
	71	1915	71000	51 0½	2	GE-259-A	}
	168	1917	75500	51 0½	2	GE-260-A	}
Metropolitan West Side Elev. Rwy.	12	1895-98	65000	47 9½	2	GE-2000	}
	78	1904	65-70000	47 9½	2	GE-55	} Yes
	37	1914-5	70000	48 6¾	2	GE-243	}

Northwestern Elev. R. R.	192	1900-06	65-69000	46 7½	2	GE-55	}
	20	1908	69000	46 7½	2	GE-211-B	} Yes
	43	1914-15	70000	48 6¾	2	GE-243	}
South Side Elev. R. R.	149	1897	52714	47 1	2	GE-57-B	}
	70	1902-03	55000	47 3	2	GE-73-A	} Yes
	61	1914-15	70000	48 6¾	2	GE-243	}
Chicago & Oak Park Elev. R. R.	84	1906	65-70000	46 7½	2	GE-55	Yes
N. Y. Municipal Ry.	900	1914-20	90600	67 0	2	GE-248	Yes
Philadelphia Rapid Tran. Co.	40	1906	71000	49 7	2	GE-66	}
	40	1907	76000	49 7	2	GE-66	}
	16	1906	72000	49 7	2	GE-66	}
	4	1909	76000	49 7	2	GE-66	} No
	65	1911	70500	49 7	2	GE-66	}
	50	1913	70000	49 7	2	GE-222-	}

						F	
	50	1921	89600	55	2	GE-259	}
	50	1922	89600	55	2	GE-259	}

Bulletin Number 49 is a reproduction of a 1922 General Electric Company pamphlet. Since that year many changes have been made in the systems described and new lines have been constructed in Cleveland, Toronto and Montreal. Another is under construction in the San Francisco area. Additional copies are available at $1.50 each from the Electric Railway Historical Society, Box 3305, Chicago, Ill. 60654.

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